

Supplementary Information

Coverage-Induced Agreement Inflation in Higher-Order Representations of Sparse Affiliation Networks

Supplementary Table S1. Chain-level APPLIES $q = 2$ degree-preserving fixed-margin audit.

All four chains started from the observed APPLIES bipartite graph. Burn-in, sampling intervals, and totals are proposal attempts; rejected or invalid proposals were counted and treated as self-loops. Each retained graph was checked for exact preservation of all labeled entity degrees, labeled context sizes, the total edge count, and matrix dimensions, as well as satisfaction of the binary-incidence constraint. Because entity degrees were preserved, the APPLIES-active population was unchanged.

Panel A. Overall fixed-margin audit results and diagnostic summary

Audit quantity	Value
Observed statistic, ρ_{obs}	0.7154799251
Pooled retained-sample mean	0.5709541982
Pooled retained-sample population SD	0.0887987450
Pooled quantiles (2.5%, 50%, 97.5%)	0.3765862222; 0.5766076104; 0.7287877984
Mean-centered two-sided tail count / retained graphs	421 / 4,000
Mean-centered two-sided empirical tail proportion	0.1052500000
Approximate plus-one two-sided p_{MC}	0.1054736316
Directional upper-tail count / retained graphs	153 / 4,000
Directional upper-tail empirical proportion	0.0382500000
Approximate plus-one upper-tail p_+	0.0384903774
Split- $\hat{R}$ for ρ_r	1.000000
Minimum per-chain ESS for ρ_r	525.6299
Minimum per-chain ESS for tail indicator	872.0225
Minimum per-chain ESS for upper-tail indicator I_r^+	881.8188
Exact labeled-margin preservation	PASS (4,000 / 4,000 retained graphs)
20-point burn-in trace review	no sustained common-direction trend
Predefined 200 E burn-in extension triggered	No

Panel B. Chain configuration and proposal accounting

Chain	Seed	Burn-in	Interval	Retained	Sampling proposals	Total proposals	Accepted switches	Acceptance rate
1	2026080501	15,300	153	1,000	153,000	168,300	101,497	0.603072
2	2026080502	15,300	153	1,000	153,000	168,300	101,651	0.603987
3	2026080503	15,300	153	1,000	153,000	168,300	101,705	0.604308
4	2026080504	15,300	153	1,000	153,000	168,300	101,822	0.605003

Panel C. Chain-specific retained-sample distributions

Chain	Retained-sample mean	Population SD	2.5% quantile	Median	97.5% quantile
1	0.5718935535	0.0902952147	0.3721988405	0.5790641641	0.7304297811
2	0.5695559627	0.0887928054	0.3842891158	0.5737344820	0.7289515421
3	0.5731813948	0.0852592573	0.3834581102	0.5781138161	0.7227833570
4	0.5691858818	0.0906830449	0.3771482750	0.5730781476	0.7294095996

Panel D1. Chain-specific mean-centered two-sided tail diagnostics

Chain	Tail count	Tail frequency	ESS, ρ_r	ESS, I_r	MCSE, tail frequency	Margins
1	117	0.117000	624.9518	872.0225	0.010885	PASS
2	100	0.100000	634.6125	884.0864	0.010090	PASS
3	91	0.091000	525.6299	1000.0000	0.009095	PASS
4	113	0.113000	688.4071	902.3448	0.010539	PASS

Panel D2. Chain-specific directional upper-tail diagnostics

Chain	Upper-tail count	Upper-tail frequency	ESS, I_r	MCSE
1	45	0.045000	881.8188	0.006981
2	33	0.033000	951.1910	0.005792
3	37	0.037000	1000.0000	0.005969
4	38	0.038000	896.6103	0.006385

Notes. The audited statistic is the midrank Spearman correlation between K and P_2 over the fixed APPLIES-active population. Population SD uses a denominator of 4,000 for the pooled distribution in Panel A and 1,000 for each chain in Panel C. Here, $R = 4,000$.

For the mean-centered two-sided diagnostic, tail membership is defined by

$$|\rho_r - \bar{\rho}| \geq |\rho_{\text{obs}} - \bar{\rho}|,$$

where $\bar{\rho}$ is the pooled retained-sample mean, and

$$pMC = (1 + \text{two-sided tail count}) / (R + 1).$$

For the directional upper-tail sensitivity diagnostic, tail membership is defined by

$$\rho_r \geq \rho_{\text{obs}},$$

and

$$p_+ = (1 + \text{upper-tail count}) / (R + 1).$$

Both quantities are interpreted as approximate MCMC tail diagnostics rather than exact randomization-test p-values. All four chains in the primary audit were initialized from the observed APPLIES bipartite graph; accordingly, its between-chain diagnostics are interpreted as practical finite-run stability checks after burn-in rather than as evidence from dispersed initializations. A separate dispersed-start initialization-sensitivity analysis is reported below in Supplementary Table S2.

ESS values use the initial positive sequence estimator. For each binary tail-indicator sequence, MCSE is calculated as $\text{MCSE}(\hat{p}_j) = \sqrt{[\hat{p}_j(1 - \hat{p}_j)/\text{ESS}_j]}$.

Acceptance rates are reported descriptively rather than used as convergence criteria.

Software and source records. The fixed-margin audit was implemented using `audit_null_model_chains_v3.py`, version 3.0.0-20260806, under CPython 3.12.10, with Python's `random.Random` implementation of the MT19937 generator.

Alternative-initialization sensitivity analysis

As an additional initialization-sensitivity check, the four APPLIES $q = 2$ fixed-margin switch chains were re-run from independently randomized fixed-margin starting realizations rather than directly from the observed APPLIES graph. Each starting realization was generated using $1000|E|$ proposal attempts under the same degree-preserving switch mechanism. All labeled entity degrees, labeled context sizes, the binary-incidence constraint, and the total edge count were preserved exactly.

The four randomized starting realizations had edge-set symmetric differences of 218, 216, 244, and 220 from the observed graph, respectively; pairwise distances among the four starting realizations ranged from 216 to 250. After initialization, the formal audit retained the same $100|E|$ burn-in, $|E|$ sampling interval, and 1,000 retained samples per chain used in the primary audit.

Across the resulting 4,000 retained graphs, the pooled mean Spearman association was 0.5726842, with empirical 2.5%, 50%, and 97.5% quantiles of 0.3865995, 0.5790241, and 0.7313256. The mean-centered two-sided tail count was 421 of 4,000 retained graphs, yielding $p_{MC} = 0.1054736$, identical to the primary same-start audit at the displayed precision. The directional upper-tail count was 168 of 4,000, yielding $p_{+} = 0.0422394$.

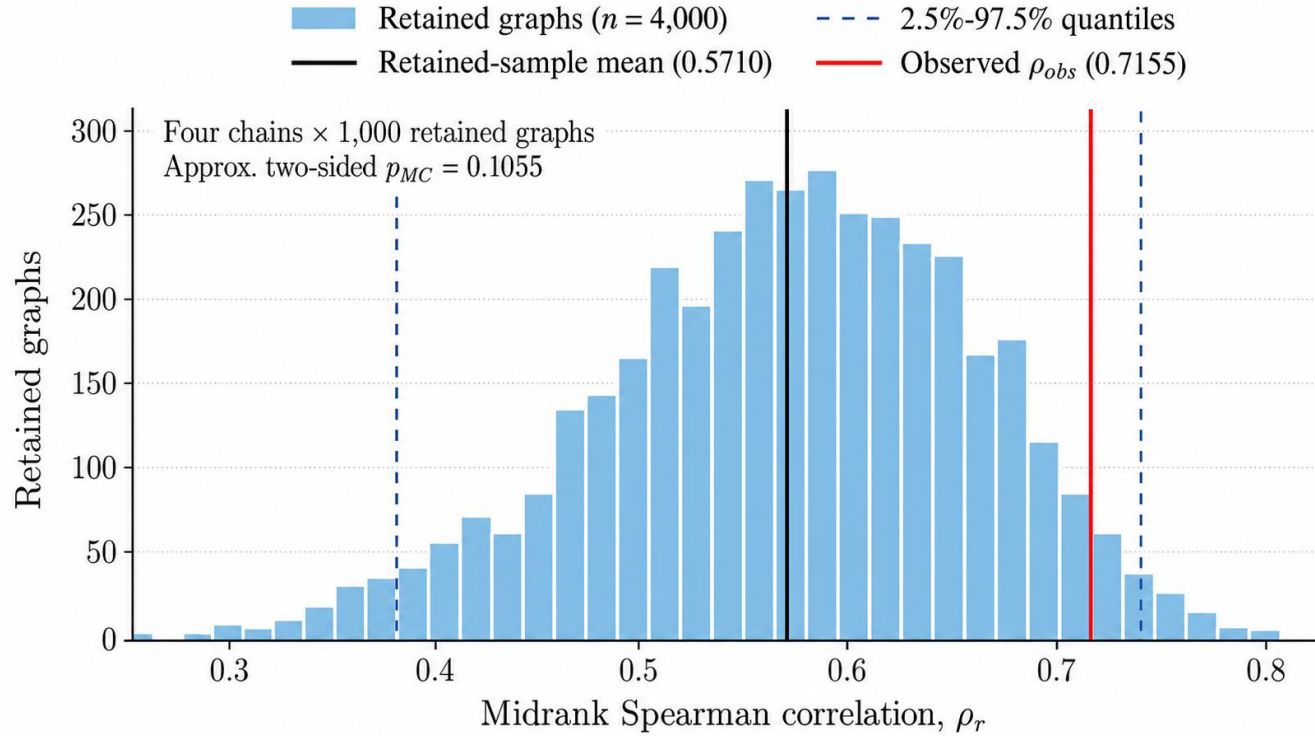
The split- $\hat{R}$ for ρ_r was 1.000314 and the minimum per-chain effective sample size for ρ_r was 601.22. The retained-sample means across the four chains ranged from 0.56998 to 0.57435. Visual review of the burn-in traces showed no sustained common-direction trend across chains. Taken together, these results indicate that the substantive fixed-margin interpretation is insensitive to the tested initialization scheme. This additional analysis does not establish exact finite-run convergence, independence of retained states, or uniform sampling over the complete fixed-margin state space.

Supplementary Table S2. Sensitivity of the APPLIES $q = 2$ fixed-margin audit to chain initialization.

Initialization	Mean ρ_r	2.5%	Median	97.5%	p_{MC}	p_{+}	Split- $\hat{R}$	Min. ESS
Observed-graph start	0.5709542	0.3765862	0.5766076	0.7287878	0.1054736	0.0384904	1.000000	525.63
Dispersed starts	0.5726842	0.3865995	0.5790241	0.7313256	0.1054736	0.0422394	1.000314	601.22

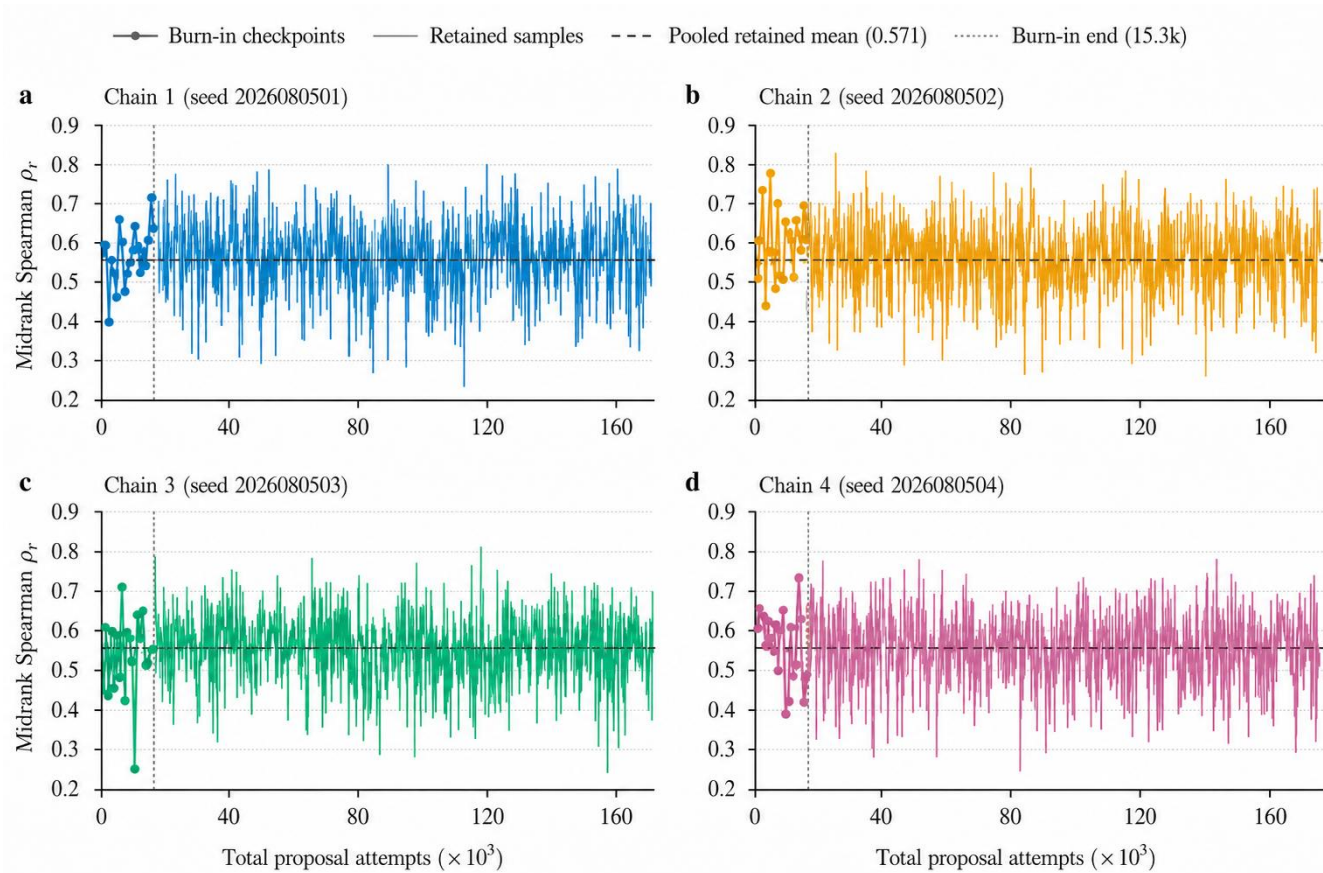
Notes. The dispersed-start audit is an additional initialization-sensitivity analysis and does not replace the primary fixed-margin audit reported in Supplementary Table S1. Each dispersed starting realization was generated using $1000|E|$ proposal attempts before the formal $100|E|$ burn-in. The same retention and diagnostic definitions were used in both analyses.

Additional-audit software and source records. The dispersed-start sensitivity audit was implemented using `audit_null_model_dispersed_starts_v1.py`, version 1.0.0-20260902, under CPython 3.13.2 with Python's `random.Random` implementation of the MT19937 generator. The run recorded SHA-256 hashes for the dispersed-start script, the primary audit script, the switch implementation, the representation diagnostic module, and the APPLIES input file.



Supplementary Fig. S1 | Degree-preserving fixed-margin reference distribution of the APPLIES active-set correlation. The histogram shows the midrank Spearman correlations obtained from 4,000 retained degree-preserving bipartite graphs across four chains, with 1,000 retained graphs per chain. All chains were initialized from the observed APPLIES graph, and every labeled entity degree, every labeled context size, and the total edge count were preserved exactly. The solid black line marks the pooled retained-sample mean (0.5710), the dashed blue lines mark the empirical 2.5% and 97.5% quantiles (0.3766 and 0.7288), and the red line marks the observed statistic (0.7155). The mean-centered two-sided tail criterion yields $p_{MC} = 0.1055$. Under the directional upper-tail criterion, 153 of the 4,000 retained graphs have $\rho_r \geq \rho_{obs}$, yielding $p_+ = 0.0385$. The two quantities address different reference questions and are interpreted as approximate MCMC diagnostics rather than exact randomization-test p -values.

Notes. The audited statistic is the midrank Spearman correlation between K and P_2 over the fixed APPLIES-active population. Here, $R = 4,000$, and tail membership is defined by $|\rho_r - \bar{\rho}| \geq |\rho_{obs} - \bar{\rho}|$, where $\bar{\rho}$ is the pooled retained-sample mean. The reported $p_{MC} = (1 + \text{tail count})/(R + 1)$ is interpreted as an approximate mean-centered two-sided MCMC tail diagnostic rather than an exact randomization-test p -value. All four chains represented in this figure were initialized from the observed APPLIES bipartite graph; accordingly, the between-chain diagnostics shown here are practical stability checks after burn-in. A separate dispersed-start initialization-sensitivity analysis is reported in Supplementary Table S2. ESS values use the initial positive sequence estimator, MCSE is calculated as $\sqrt{[\hat{p}_j(1 - \hat{p}_j)/\text{ESS}_j]}$, and acceptance rates are reported descriptively rather than used as convergence criteria.



Supplementary Fig. S2 | Four-chain burn-in and retained-sample traces for the APPLIES $q = 2$ fixed-margin audit. a–d, Chain-specific trajectories of the midrank Spearman correlation for seeds 2026080501, 2026080502, 2026080503, and 2026080504. All four chains were initialized from the observed APPLIES bipartite graph. Filled circles show 20 equally spaced burn-in checkpoints over 15,300 proposal attempts, and thin lines show all 1,000 retained samples per chain, recorded every 153 proposal attempts during the subsequent 153,000-proposal sampling phase. The vertical dotted line marks the end of burn-in, and the horizontal dashed line marks the pooled retained-sample mean (0.5709542). The trajectories fluctuate around the retained-sample center without a sustained common-direction trend across chains. This visual assessment is interpreted together with the Split- $\hat{R}$ of 1.000000, the minimum per-chain correlation-statistic effective sample size of 525.6299, the minimum mean-centered two-sided tail-indicator effective sample size of 872.0225, and the minimum directional upper-tail indicator effective sample size of 881.8188; it does not establish exact finite-run convergence or independence of the retained states. Source data are provided in Supplementary_Fig_S2_trace_source_data.csv.